Modular Arithmetic, Divisibility, Primes & Euler Totient Practice Exercises

1 Using Legendre's prime-valuation theorem, how many trailing zeros appear at the end of $35!$ in base 10?

a. 7

b. 9

c. 8

d. 6

2 Using Legendre's prime-valuation theorem, how many trailing zeros appear at the end of $50!$ in base 10?

a. 10

b. 12

c. 13

d. 8

3 Using Legendre's prime-valuation theorem, how many trailing zeros appear at the end of $65!$ in base 10?

a. 15

b. 13

c. 16

d. 11

4 Using Legendre's prime-valuation theorem, how many trailing zeros appear at the end of $80!$ in base 10?

a. 16

b. 20

c. 17

d. 19

5 Let x be the smallest positive integer satisfying the simultaneous congruences $x \equiv 3 \pmod 5$ (whose initial progression is marked on the number line) and $x \equiv 6 \pmod 7$. Find x.

a. 18

b. 20

c. 13

d. 22

© ClipSAT — clipsat.org | All Rights Reserved

6 Let x be the smallest positive integer satisfying the simultaneous congruences $x \equiv 1 \pmod 5$ (whose initial progression is marked on the number line) and $x \equiv 7 \pmod 9$. Find x.

a. 21

b. 16

c. 25

d. 29

7 Let x be the smallest positive integer satisfying the simultaneous congruences $x \equiv 5 \pmod 7$ (whose initial progression is marked on the number line) and $x \equiv 1 \pmod 9$. Find x.

a. 19

b. 26

c. 28

d. 44

8 Let x be the smallest positive integer satisfying the simultaneous congruences $x \equiv 1 \pmod 7$ (whose initial progression is marked on the number line) and $x \equiv 0 \pmod{11}$. Find x.

a. 29

b. 33

c. 55

d. 22

9 Given that $p = 11$ is prime, find the least non-negative remainder when $9!$ is divided by 11.

a. 10

b. 2

c. 1

d. -3

10 Given that $p = 13$ is prime, find the least non-negative remainder when $10!$ is divided by 13.

a. 7

b. 6

c. 12

d. 2

11 Given that $p = 17$ is prime, find the least non-negative remainder when $13!$ is divided by 17.

a. 3

b. 14

c. 4

d. 16

© ClipSAT — clipsat.org | All Rights Reserved

12 Given that $p = 19$ is prime, find the least non-negative remainder when $17!$ is divided by 19.

a. 18

b. 2

c. -3

d. 1

© ClipSAT — clipsat.org | All Rights Reserved